

Software Requirement Specification

Generated & Accepted: 9/8/2026, 6:04:44 PM | Total Requirements: 7

REQ-001 [REQ-001] CLI Conversational Interface & Session Lifecycle

Summary

The system provides a single command-line interface (CLI) for users to interactively submit natural-language queries, maintain session history during the conversation, and terminate the session using exit commands.

Acceptance Criteria

- ☐ The system shall provide a single CLI interface through which a user can submit natural-language questions.
- ☐ The user can terminate the session using exit or quit.
- ☐ The conversation history shall be updated with the user request and assistant response during the current CLI session.

Assumptions

- ☐ Persistent conversation storage, graphical user interface, and user authentication are not supported.

Dependencies

- ☐ Interactive command-line execution environment (agent.py).

Edge Cases

- ☐ User terminates the session explicitly using 'exit' or 'quit' commands.

REQ-002 [REQ-002] Natural Language Request Classification and Routing

Summary

The system automatically classifies incoming natural-language queries into one of four distinct processing routes (A, B, C, or D) to trigger specialized capabilities or general language model responses.

Acceptance Criteria

- ☐ The system shall automatically classify each request into an appropriate processing route (A, B, C, or D).

Assumptions

- ☐ Classification relies on query intent matching the four supported routes: date/time, calculation, web search, and general QA.

Dependencies

- ☐ Gemini model service and agent classification logic in agent.py.

Edge Cases

- ☐ Requests that do not match specialized routes A, B, or C are routed directly to route D (Gemini general QA).

REQ-003 [REQ-003] Current Date and Time Retrieval

Summary

The system routes current date and time inquiries to a dedicated local date/time utility to return accurate, grounded temporal information.

Acceptance Criteria

- ☐ A current-date/time request is routed to the date/time capability.

â€¢ Tool output is used as the basis for tool-derived answers and shall not intentionally introduce unsupported facts.

Assumptions

â€¢ The date/time capability is executed via local system time utilities.

Dependencies

â€¢ Local date/time service / tools.py datetime utility.

Edge Cases

â€¢ Failure in date/time execution is caught and reported rather than fabricating a result.

REQ-004 â€” [REQ-004] Mathematical Calculation Engine

Summary

The system routes mathematical calculation requests to a dedicated SymPy-based calculation engine rather than relying on the language model to perform arithmetic or symbolic evaluations.

Acceptance Criteria

â€¢ A mathematical request is routed to the calculator capability.

â€¢ The system shall use a dedicated calculator capability for mathematical calculations rather than relying on the language model to perform the calculation.

â€¢ Tool output is used as the basis for tool-derived answers.

Assumptions

â€¢ The calculation capability relies on SymPy to parse and evaluate mathematical expressions.

Dependencies

â€¢ SymPy calculation engine (tools.py / requirements.txt).

Edge Cases

â€¢ Invalid mathematical expressions or evaluation errors are caught and reported without silent fabrication.

REQ-005 â€” [REQ-005] Web Search and Current Information Research

Summary

The system routes requests requiring real-time, current, or internet-dependent information (such as news, weather, stock prices, sports, companies, or politics) to the Tavily web search service and synthesizes grounded, structured responses.

Acceptance Criteria

â€¢ A current-information/web-research request is routed to web search.

â€¢ The research prompt instructs the model to prefer official sources, compare multiple sources when possible, identify conflicts, and state when information cannot be verified.

â€¢ Web-search responses are structured with headings and may include bullets, tables, references, statistics, examples, comparisons, and future trends when applicable.

â€¢ Tool-derived answers shall be grounded in the returned tool output and shall not intentionally introduce unsupported facts.

Assumptions

â€¢ Tavily API requires valid network access and API credentials provided via environment configuration.

Dependencies

- â€¢ Tavily web-search service and API key configuration.
- â€¢ Gemini model service for synthesizing research answers.

Edge Cases

- â€¢ When web information cannot be verified or sources conflict, the system states that the information cannot be verified or identifies the conflict.
- â€¢ Tavily tool failures or search errors are reported without fabricating a successful result.

REQ-006 â€” [REQ-006] Direct General Question Answering

Summary

The system processes general queries that do not require external tools or calculation directly through the Gemini model, incorporating conversation history to maintain context.

Acceptance Criteria

- â€¢ A general request can be answered directly by Gemini.
- â€¢ Requests that do not require route A, B, or C shall be sent directly to the Gemini model with the session history.
- â€¢ The system shall return a natural-language answer to the user.

Assumptions

- â€¢ Direct question answering applies only within the ongoing active CLI session without persistent long-term storage.

Dependencies

- â€¢ Gemini model service and API credentials.

Edge Cases

- â€¢ API communication or model generation runtime errors are caught and surfaced to the user.

REQ-007 â€” [REQ-007] Configuration & Runtime Error Handling

Summary

The system validates environment configurations on startup and catches runtime or tool execution errors gracefully, displaying controlled error messages without fabricating output.

Acceptance Criteria

- â€¢ Missing GEMINI_API_KEY shall prevent startup and produce a configuration error.
- â€¢ Exceptions during route execution shall be caught by the main interaction loop and displayed as an error.
- â€¢ A tool failure is reported without silently fabricating a successful result.

Assumptions

- â€¢ Environment configuration is loaded from .env or system environment variables.

Dependencies

- â€¢ Environment variables configuration (.env file with GEMINI_API_KEY and Tavily keys).

Edge Cases

- â€¢ Invalid or missing API keys prevent startup and output a configuration error.
- â€¢ Unhandled exceptions occurring during any route execution are caught by the loop and displayed rather than terminating unexpectedly.